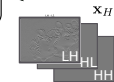




Haar
Wavelet
Transform



Decoupled Bands
 $\mathbf{x}_L, \mathbf{x}_H$



$\mathbf{x}_L$

Low Freq.
Encoder

$$\mathcal{L}_{\text{rec}}^L = \|\mathbf{x}_L - \hat{\mathbf{x}}_L\|_2^2$$



$$\tilde{\mathbf{z}} = [\mathbf{z}_L \parallel \mathbf{z}_H]$$

Low Freq.
Decoder

$\hat{\mathbf{x}}_L$



$\mathcal{L}^L$

Inverse
Wavelet
Transform



High Freq.
Encoder

$$\mathcal{L}_{\text{rec}}^H = \|\mathbf{x}_H - \hat{\mathbf{x}}_H\|_2^2$$



High Freq.
Decoder

$\hat{\mathbf{x}}_H$



$\mathcal{L}^H$

Decoupled Bands $\hat{\mathbf{x}}_L, \hat{\mathbf{x}}_H$
with Separate Losses $\mathcal{L}^L, \mathcal{L}^H$



$$\tilde{\mathbf{z}} \sim p_{\theta}(\tilde{\mathbf{z}})$$

Latent Diffusion

$$\mathcal{L}^L = \mathcal{L}_{\text{rec}}^L + \beta \mathcal{L}_{\text{KL}}^L + \lambda_{\text{VF}} \mathcal{L}_{\text{VF}}^L + \lambda_{\text{GAN}} \mathcal{L}_{\text{GAN}}^L + \lambda_{\text{LPIPS}} \mathcal{L}_{\text{LPIPS}}^L$$

$$\mathcal{L}^H = \mathcal{L}_{\text{rec}}^H + \beta \cdot \mathcal{L}_{\text{KL}}^H + \mathcal{L}_{\text{GAN}}^H$$